

MODULE-01

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Foundations of probability.

- * probability is the mathematics of uncertainty
- * It measures the likelihood of an event occurring.
- * values of probability lie between 0 & 1.

Applications.

1. Weather Forecasting
 - (When you hear "There is 70% chance of rain tomorrow", it means:
out of many days in the past with similar weather conditions, it rained on 70% of those days.
 - Meteorologists use probability based on past data & computer models.)
 - If similar weather conditions occurred 100 times in the past and it rained on 70 of those days
probability of rain tomorrow = 70%.
2. Medical Testing
 - A test with 95% accuracy means: out of 100 sick patients, 95 test positive. Doctors use conditional probability (Bayes' theorem) to interpret results.
3. Finance & Insurance
 - If chance of accident = 0.02 (2%) & average loss = ₹ 50,000
 - Expected Cost = $0.02 \times 50,000 = ₹ 1000$.
 - Premium is set slightly higher.

Introduction to probability.

It is a mathematical framework for quantifying uncertainty. It measures how likely an event is to occur, ranging from 0 (impossible event) to 1 (certain event).

"probability" is the chance that something will happen.

ex:- Getting a sun³ when tossing a coin. $\rightarrow$ Chance = 0.
impossible event.

Getting a head when tossing a coin $\rightarrow$ Chance = 0.5
possible event.

Basic probability concepts.

* Random experiments:

It is a process or action which can be repeated under the same conditions. Has two or more possible outcomes and the outcome cannot be predicted with certainty in advance.

ex:- Tossing a coin or rolling a die.

* Sample Space:

The set of all possible outcomes of a random experiment is called Sample Space.

It is denoted by 'S'.

ex:- 1. Tossing a coin (Random experiment)

$$S = \{H, T\} \text{ (Sample Space)}$$

2. Rolling a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

3. Tossing two coins

$$S = \{HH, HT, TH, TT\}$$

4. Tossing three coins

$$S = \{HHH, HHT, HTH, HTT, TTT, TTH, THT, TTH\}$$

5. Rolling two dice.

$$S = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

* Sample point

Each element of a sample space are called sample point. (Each individual outcome)

Ex: Tossing the coin

Sample Space $S = \{H, T\}$

Sample points H or T

* Event

The outcome of an random experiment is called an event.

An event is any subset of the sample space of a random experiment.

It is denoted by 'E'

i.e $E \subset S$

Ex: Rolling a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Event: "Getting an even number"

$E = \{2, 4, 6\}$ Here E is the subset of S

Classical definition of probability.

Let S be the sample space then the probability of occurrence of an event E defined as

$$P(E) = \frac{\text{no. of elements in the event } E}{\text{no. of elements in the sample space } S}$$

$$P(E) = \frac{n(E)}{n(S)}$$

#

operation of events | Types of events

Let S be a sample space, and $A, B \subseteq S$ be two events

1. Union of Events

$$A \cup B = \{x \in S \mid x \in A \text{ or } x \in B\}$$

2. Intersection of Events

$$A \cap B = \{x \in S \mid x \in A \text{ \& } x \in B\}$$

3. Complement of an Event

$$A' = \{x \in S \mid x \notin A\} = S - A$$

4. Impossible Event

$$\phi = \{\}$$

union and Intersection of Events

Let A and B are two events in a sample space S , then the union of A & B is an event, denoted by $A \cup B$ is defined as

$$A \cup B = \{x \in S \mid x \in A \text{ or } x \in B\}$$

and the intersection of A and B is an event, denoted by $A \cap B$ is defined as

$$A \cap B = \{x \in S \mid x \in A \text{ and } x \in B\}$$

$A \cup B$ is also referred to as "event A or B", & $A \cap B$ is referred to as "event A & B".

Complement of an Event.

Let S be the sample space for a random experiment and E be an event then the complement of 'E' is denoted by E^c or E' or $\bar{E}$ which represents the event where E does not occur.

Ex:- Rolling a die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let A be the event of getting odd numbers

$$A = \{1, 3, 5\}$$

Let B be the event of getting even number

$$B = \{2, 4, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$A \cap B = \{\phi\}$$

$$A' = \{2, 4, 6\}$$

$$B' = \{1, 3, 5\}$$

Types of Events

1. Simple event - An event having only one sample point
Ex:- If we tosses two coins

$$E_1 = \{HH\} \quad E_2 = \{TT\}$$

2. Compound event - An event which is not a simple event.

Ex:- If we tosses two coins

$$E_1 = \{HH, HT, TH\} \quad E_2 = \{TH, HT\}$$

3. Mutually Exclusive Events

If two or more events cannot occur simultaneously then they are called mutually exclusive or disjoint events.

Thus two events A & B are mutually exclusive if $A \cap B = \phi$.

In general a set of events $E_1, E_2, \dots, E_n$ are mutually exclusive if $E_1 \cap E_2 \cap E_3 \dots = \phi$

Ex:- 1. If we throw an unbiased die

$$A = \{2, 4, 6\} \quad \& \quad B = \{1, 3, 5\} \quad A \cap B = \phi$$

then A & B are mutually exclusive events.

2. A = Event of getting a prime number
B = Event of getting an odd number

$$A = \{3, 5\} \quad B = \{1, 3, 5\}$$

$$A \cap B \neq \phi$$

A & B are not mutually exclusive events.

3. Exam result $\rightarrow$ pass or fail.

4. Medical test $\rightarrow$ positive or negative

4. Exhaustive event
A set of events is said to be exhaustive if it includes all the possible events.

A set of events $E_1, E_2, \dots, E_n$ are said to be exhaustive if $E_1 \cup E_2 \cup \dots \cup E_n = S$ where S is a sample space.

Ex: 1. If we throw a die then $S = \{1, 2, 3, 4, 5, 6\}$
 $E_1 = \{2, 4, 6\}$ & $E_2 = \{1, 3, 5\}$
 $E_1 \cup E_2 = S$
 E_1 & E_2 are exhaustive events.

• Days of a week - $\{M, T, W, Th, F, S, S\}$
 • Weather - $\{\text{rain, no rain}\}$
 • Traffic signal - $\{\text{Red, yellow, green}\}$
 $E_1 = \{1, 2, 3\}$ $E_2 = \{5, 6\}$
 $E_1 \cup E_2 = \{1, 2, 3, 5, 6\}$
 E_1 & E_2 are not exhaustive events.

• $E_1 = \{1, 2, 3, 4\}$ $E_2 = \{3, 4, 5, 6\}$
 $E_1 \cup E_2 = \{1, 2, 3, 4, 5, 6\} = S$
 E_1 & E_2 are exhaustive events.

Note :- The event A & A' are mutually exclusive & exhaustive.

Addition theorem of mutually exclusive events
 Statement - let A & B are the two events then the probability of occurrence of at least one of the events A & B is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A & B are mutually exclusive events then $A \cap B = \phi$

$$\therefore P(A \cup B) = P(A) + P(B)$$

[When we add $P(A)$ & $P(B)$ the common part is counted twice, we subtract $P(A \cap B)$ to correct it.]

Problems:-

1. If Two dice are thrown simultaneously then find the probability of following events & verify if they are exhaustive & mutually exclusive.

A: The sum of the numbers is 7

B: The sum of the numbers is not 7.

Soln

$$S = \left\{ \begin{array}{l} (1,1) (1,2) (1,3) (1,4) (1,5) (1,6) \\ (2,1) (2,2) (2,3) (2,4) (2,5) (2,6) \\ (3,1) (3,2) (3,3) (3,4) (3,5) (3,6) \\ (4,1) (4,2) (4,3) (4,4) (4,5) (4,6) \\ (5,1) (5,2) (5,3) (5,4) (5,5) (5,6) \\ (6,1) (6,2) (6,3) (6,4) (6,5) (6,6) \end{array} \right\} \quad n(S) = 36$$

i). Event A - Sum of the numbers is 7

$$A = \{ (1,6) (2,5) (3,4) (4,3) (5,2) (6,1) \}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

ii) Event B - Sum of the numbers is not 7

$$\begin{aligned} n(B) &= n(S) - n(A) \\ &= 36 - 6 \\ &= 30 \end{aligned}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{30}{36} = \frac{5}{6}$$

$(A \cup B)$ covers the entire sample space 'S' & (E)

$(A \cap B)$ is empty set. (M.E)

$\therefore A$ & B are mutually exclusive & exhaustive.

2. A bag contains 3 red balls & 2 black balls. A ball is drawn at random. Let event A be the ball drawn is red & event B is the ball drawn is black. Verify A & B are mutually Exclusive & Exhaustive.

Soln

Total balls = 5

Event A: ball is red

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{5}$$

Event B: ball is black

$$P(B) = \frac{n(B)}{n(S)} = \frac{2}{5}$$

$A \cup B = S$ Events are Exhaustive.

$A \cap B = \emptyset$ Events are mutually exclusive.

Approaches to probability

1) Classical probability Approach

when there are 'n' equally likely outcomes to an experiment $P(A) = \frac{\text{no of favorable outcomes}}{\text{Total number of outcomes}} = \frac{n}{N}$

Ex:- 1. The chance that tail or head appears in a toss of a coin. $P = \frac{1}{2}$ [fair die, fair coin, well shuffled cards]

2. Rolling a die.

$$P(\text{Even}) = \frac{3}{6} = \frac{1}{2}$$

(or)

[when outcomes are finite, equally likely & known in advance]

Let S be the Sample Space of an experiment, & let E be a particular outcome of the experiment. Then the probability of event E is denoted by $P(E)$.

② Relative Frequency probability Approach

If an experiment is repeated N times & an event E occurs f times, then the probability of E is given by $P(E) = \frac{f}{N}$

$$P(E) = \frac{\text{no. of times } E \text{ occur}}{\text{total no. of trials}}$$

Ex: A coin is tossed 50 times $\rightarrow$ Head appears 28 times.

$$P(\text{Head}) = \frac{28}{50} = 0.56$$

Ex: weather, traffic,

class performance $\Rightarrow$ 35 students
50 Total students

* Axiomatic approach definition of probability

Let S be the Sample Space & A be an event associated with the random experiment then the probability of the event A denoted by $P(A)$ is defined as a real number satisfying the following axioms.

- i) $0 \leq P(A) \leq 1$
- ii) $P(S) = 1$
- iii) If A & B are mutually exclusive events
 $P(A \cup B) = P(A) + P(B)$
- iv) If $A_1, A_2, \dots, A_n$ are set of mutually exclusive events
 $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

1. A coin is tossed 40 times & heads appear 22 times. Find the relative frequency of getting heads.

Solⁿ:- A coin is tossed 40 times & head appears 22 times.

using the relative frequency approach:

$$P(\text{Heads}) = \frac{\text{number of times heads occurred}}{\text{Total tosses}}$$

$$= \frac{22}{40} = 0.55$$

So, the probability of getting heads is 0.55.

A student throws a die 50 times. The number 6 appeared 9 times. Find the relative frequency of getting a 6.

Solⁿ The die was thrown 50 times & the number 6 appeared 9 times.

using the relative frequency approach

$$P(6) = \frac{\text{number of times 6 appeared}}{\text{Total trials}}$$

$$= \frac{9}{50} = 0.18$$

So, the probability of getting a 6 is 0.18.

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- (3) A bag contains 4 red & 6 black balls. One ball is drawn at random. Find the classical probability that it is
a) red
b) not red.

Soln: The total number of balls is 4 red + 6 black = 10.

a) probability of drawing a red ball

The favourable outcomes = 4

Total possible outcomes = 10

$$P(\text{Red}) = \frac{\text{No. of red balls}}{\text{Total balls}} = \frac{4}{10} = 0.4$$

b) probability of drawing a ball that is not red (black):

The favourable outcomes = 6

Total possible outcomes = 10

$$P(\text{not red}) = \frac{\text{no. of black balls}}{\text{Total balls}} = \frac{6}{10} = 0.6$$

So, the classical probabilities are:

red = 0.4, not red = 0.6.

4. Standard Deck of cards.
It has 52 cards divided into 4 suits:
Spades, Hearts, Clubs, Diamonds
(13) (13) (13) (13)

no: 2, 3, 4, 5, 6, 7, 8, 9, 10

face cards: Jack(J), Queen(Q),
King(K)

• Ace(A).

4.

A card is drawn from a deck. find the probability (classical) that the card is:

a) A Spade

b) A number less than 5

c) Not a red card.

Soln: The classical frequency formula is:

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes.}}$$

a) A Spade: There are 13 Spade cards in the deck
 $P(\text{spade}) = \frac{13}{52} = 0.25$

There is a 25% chance of drawing a Spade from a Standard deck.

b) A number less than 5:
 There are 4 cards with number less than 5 in all four suits.

Ace(A), 2, 3, 4.
 So, Total such cards = $4 \times 4 = 16$.

$$P(\text{A number less than 5}) = \frac{16}{52} = 0.31$$

There is a 31% chance of drawing a card number less than 5 is about 31%.

c) Not a red card:

Hearts & diamonds - red (26)

Spades & clubs - Black (26).

$$P(\text{not red}) = \frac{26}{52} = 0.5$$

There is a 50% chance of drawing a card that is not red.

5- using Axiomatic approach, prove

a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

b) $P(A^c) = 1 - P(A)$

c) $P(\phi) = 0$.

Solⁿ: a). Let A & B be two events in a sample space S

- The event $A \cup B$ means "either event A or event B or both occurs".
- When we add $P(A) + P(B)$, we are also including the outcomes common to both A & B twice (i.e. outcomes in $A \cap B$).
- So, we must subtract $P(A \cap B)$ once to avoid counting it twice.

$$\text{Hence } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This is called "addition rule of probability."

b) The complement of event A includes all outcomes not in A . Since A & its complement together make the entire sample space & they have no common elements.

$$P(A) + P(A^c) = P(S) = 1$$

$$P(A^c) = 1 - P(A)$$

c) The empty set has no outcomes, so it is an impossible event. By defⁿ

$$P(\phi) = 0.$$

Addition Rule.

Statement:- If A & B are any two events of 's' which are not mutually exclusive then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:- We prove this result using Venn diagram from the figure

$$A = (A \cap \bar{B}) \cup (A \cap B)$$

$$B = (\bar{A} \cap B) \cup (A \cap B)$$

$$\Rightarrow P(A) = P(A \cap \bar{B}) + P(A \cap B)$$

$\therefore A \cap \bar{B}$ & $A \cap B$ are disjoint

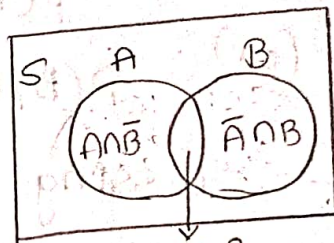
$$\text{iv. } P(B) = P(\bar{A} \cap B) + P(A \cap B)$$

$\therefore \bar{A} \cap B$ & $A \cap B$ are disjoint

now
$$P(A) + P(B) = P(A \cap \bar{B}) + P(A \cap B) + P(\bar{A} \cap B) + P(A \cap B)$$

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$

Thus
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$A \cap B$
outcomes that are in
 A but not in B

Note: If A & B are mutually exclusive then

$$P(A \cap B) = 0$$

$$\therefore P(A \cup B) = P(A) + P(B)$$

* Independence of Events

Two events A and B are said to be independent if the occurrence of one does not affect the probability of occurrence of the other.

$$P(A \cap B) = P(A) \cdot P(B)$$

$$(or) \quad P(B|A) = P(B) \quad , \quad P(A|B) = P(A)$$

Ex:- Tossing a coin & rolling a die

A $\Rightarrow$ Getting a Head on the coin

B $\Rightarrow$ Getting a 4 on the die.

$$P(A) = \frac{1}{2} \quad P(B) = \frac{1}{6}$$

$$P(A \cap B) = P(A) \cdot P(B) \\ = \frac{1}{12}$$

A & B are independent event.

* Conditional probability

The probability of an event A happening given that event B has already happened.

It is denoted by $P(A|B) = \frac{P(A \cap B)}{P(B)}$ ($P(B) > 0$)

- $P(A|B)$ - probability of A given B
- $P(A \cap B)$ - probability of both A & B happening
- $P(B)$ - probability of B happening

Multiplication Rule of probability

According to the multiplication rule of probability, the probability of occurrence of both the events A & B is equal to the product of the probability of A occurring and the conditional probability that event B occurring given event A occurs.

$$\text{i.e. } P(A \cap B) = P(A) P(B|A) \quad (or)$$

$$P(A \cap B) = P(B) P(A|B) \quad [\because P(A \cap B) = P(B \cap A)]$$

Proof:- WKT the conditional probability of event A given B is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) \neq 0$$

$$\text{Now } P(A \cap B) = P(B) P(A|B) \rightarrow (1)$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} \quad \text{where } P(A) \neq 0$$

$$\therefore P(B \cap A) = P(A) \times P(B|A)$$

$$\therefore P(A \cap B) = P(B \cap A) \quad \text{we have}$$

$$P(A \cap B) = P(A) P(B|A) \rightarrow (2)$$

from (1) & (2) we have

$$P(A \cap B) = P(A) P(B|A) = P(B) P(A|B)$$

The above result is known as the multiplication rule of probability.

Note:- If A & B are two independent events

$$\text{when } P(B|A) = P(B)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

(or) A & B are independent.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$

- $P(B|A) = \frac{P(A \cap B)}{P(A)}$

- $P(A) = P(A \cap B) + P(A \cap B')$

- $P(B) = P(A \cap B) + P(A' \cap B)$

- $P(A) + P(A') = 1$

- $P(A^c \cap B^c) = P((A \cup B)^c)$
 $= 1 - P(A \cup B)$

- $P(A \cup B)' = 1 - P(A \cup B)$

Problems.

1. Let A & B be events with $P(A) = \frac{3}{8}$ $P(B) = \frac{5}{8}$ &
 $P(A \cup B) = \frac{3}{4}$ find $P(A|B)$ & $P(B|A)$.

Solⁿ

$$P(A) = \frac{3}{8} \quad P(B) = \frac{5}{8} \quad P(A \cup B) = \frac{3}{4}$$

By addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{3}{4} = \frac{3}{8} + \frac{5}{8} - P(A \cap B)$$

$$P(A \cap B) = \frac{8}{8} - \frac{3}{4}$$

$$P(A \cap B) = 1 - \frac{3}{4} = \frac{1}{4}$$

By definition of conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{5}{8}}$$

$$= \frac{\cancel{8}}{\cancel{20}} = \frac{2}{5}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{4}}{\frac{3}{8}}$$

$$= \frac{\cancel{8}}{\cancel{12}} = \underline{\underline{\frac{2}{3}}}$$

2. Given $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$, $P(A \cup B) = \frac{1}{2}$ evaluate $P(A|B)$, $P(B|A)$, $P(A \cap B')$ & $P(A|B')$.

Soln $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$, $P(A \cup B) = \frac{1}{2}$

By additional rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{1}{2} = \frac{1}{4} + \frac{1}{3} - P(A \cap B)$$

$$P(A \cap B) = \frac{7}{12} - \frac{1}{2}$$

$$\underline{\underline{P(A \cap B) = \frac{1}{12}}}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{12}}{\frac{1}{3}} = \underline{\underline{\frac{1}{4}}}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{12}}{\frac{1}{4}} = \underline{\underline{\frac{1}{3}}}$$

$$P(A) = P(A \cap B) + P(A \cap B')$$

$$P(A \cap B') = P(A) - P(A \cap B)$$

$$= \frac{1}{4} - \frac{1}{12} = \frac{2}{12} = \underline{\underline{\frac{1}{6}}}$$

$$P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{\frac{1}{6}}{1 - P(B)} = \frac{\frac{1}{6}}{1 - \frac{1}{3}} = \frac{\frac{1}{6}}{\frac{2}{3}} = \underline{\underline{\frac{1}{4}}}$$

3. In a certain college, 25 percent of the students failed in mathematics, 15 percent failed in chemistry, & 10 percent failed in both mathematics & chemistry. A student is selected at random.
- If the student failed in chemistry, what is the probability that he or she failed in mathematics?
 - If the student failed in mathematics, what is the probability that he or she failed in chemistry?
 - What is the probability that the student failed neither in mathematics nor in chemistry?

Solⁿ

- M = Event student failed in Mathematics
- C = event student failed in chemistry
- $M \cap C$ = event students failed in both.

$$P(M) = 25\% = 0.25 \quad P(C) = 15\% = 0.15 \quad P(M \cap C) = 10\% = 0.10$$

$$i) P(M|C) = \frac{P(M \cap C)}{P(C)} = \frac{0.10}{0.15} = \frac{10}{15} = \frac{2}{3}$$

$$ii) P(C|M) = \frac{P(M \cap C)}{P(M)} = \frac{0.10}{0.25} = \frac{2}{5}$$

iii) By addition rule

$$\begin{aligned} P(M \cup C) &= P(M) + P(C) - P(M \cap C) \\ &= 0.25 + 0.15 - 0.10 \\ &= 0.30. \end{aligned}$$

students who failed neither in maths nor in chemistry

$$\begin{aligned} P(M \cup C)' &= 1 - P(M \cup C) \\ &= 1 - 0.3 \\ &= \underline{\underline{0.7}}. \end{aligned}$$

4. Let A & B be the events with $P(A) = 0.6$, $P(B) = 0.3$ +
 $P(A \cap B) = 0.2$ find i) $P(A|B)$ ii) $P(B|A)$ iii) $P(A \cup B)$
 iv) $P(A^c)$ & $P(B^c)$ v) $P(A^c|B^c)$ & $P(B^c|A^c)$.

Solⁿ

$$P(A) = 0.6$$

$$P(B) = 0.3$$

$$P(A \cap B) = 0.2$$

$$i) P(A|B) = \frac{P(A)}{P(A \cap B)} = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.3} = \frac{2}{3}$$

$$ii) P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.2}{0.6} = \frac{1}{3}$$

$$iii) P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = 0.6 + 0.3 - 0.2 = 0.7$$

$$iv) P(A^c) = 1 - P(A) = 1 - 0.6 = 0.4$$

$$P(B^c) = 1 - P(B) = 1 - 0.3 = 0.7$$

$$v) P\left(\frac{A^c}{B^c}\right) = \frac{P(A^c \cap B^c)}{P(B^c)} = \frac{0.3}{0.7} = \frac{3}{7}$$

$$P(A^c \cap B^c) = P(A \cup B)^c \\ = 1 - P(A \cup B) \\ = 1 - 0.7 \\ = 0.3$$

$$P(B^c|A^c) = \frac{P(A^c \cap B^c)}{P(A^c)} = \frac{0.3}{0.4} = \frac{3}{4}$$

Two different digits are selected at random from the digits 1 to 5.

- a) if the sum is odd what is the probability that 2 is one of the number selected?
- b) if 2 is one of the digits, what is the probability that the sum is odd?
- selecting two different digits from the set

$S = \{1, 2, 3, 4, 5\}$

number of ways $= {}^5C_2 = 10$

$$S = \{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$$

a). (condition) $A = \text{Sum is odd}$

$$A = \{(1, 2), (1, 4), (2, 3), (2, 5), (3, 4), (4, 5)\}$$

$B = 2 \text{ is included / selected.}$

$$B = \{(1, 2), (2, 3), (2, 5)\}$$

By conditional probability

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{3}{10}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{10}$$

$$P(B/A) = \frac{3}{10} \times \frac{10}{6} = \frac{1}{2}$$

ii). (condition) A - 2 is ^{digit} included one of the

$$A = \{(1,2)(2,3)(2,4)(2,5)\}$$

B - sum is odd

$$B = \{(1,2)(2,3)(2,5)\}$$

By conditional probability

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(A)}{n(S)}} = \frac{3}{10} \times \frac{10}{4}$$

$$= \underline{\underline{\frac{3}{4}}}$$

5. Two different digits are selected at random from the digits 1 through 5
- if the sum is odd what is the probability that 2 is one of the number selected?
 - if 2 is one of the digits, what is the probability that the sum is odd?

Soln

selecting two different digits from the set $\{1, 2, 3, 4, 5\}$

Number of ways $= {}^5C_2 = 10.$

$$S = \{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$$

$n(S) = 10$

let A be the event that '2' is included
 B be the event that sum is odd

$$A = \{(1, 2), (2, 3), (2, 4), (2, 5)\}$$

$$B = \{(1, 2), (1, 4), (2, 3), (2, 5), (3, 4), (4, 5)\}$$

$n(A) = 4$ $n(B) = 6$ $n(A \cap B) = 3.$

a. by conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B) / n(S)}{n(B) / n(S)}$$

$$= \frac{3/10}{6/10} = \underline{\underline{1/2}}$$

b) $\downarrow$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{n(A \cap B) / n(S)}{n(A) / n(S)}$$

$$A = \{(1, 2), (2, 3), (2, 4), (2, 5)\}$$

$n(A) = 4$

$$= \frac{3/10}{4/10} = \underline{\underline{3/4}}$$

$$B = \{(1, 2), (1, 4), (2, 3), (2, 5), (3, 4), (4, 5)\}$$

6. A die is thrown, let A be the event that the appeared number is greater than 3 & B be the event that the appeared number is even number.

find i) $P(A)$, $P(B)$, $P(A \cap B)$

ii) $P(A|B)$ & $P(B|A)$

iii) Are A & B independent events.

Soln

$$S = \{1, 2, 3, 4, 5, 6\} \quad n(S) = 6.$$

A - event no greater than 3 $A = \{4, 5, 6\} \quad n(A) = 3$

B - event even number $B = \{2, 4, 6\} \quad n(B) = 3.$

$$i) \quad P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

$$ii) \quad P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

$$iii) \quad P(B|A) = \frac{2}{3}, \quad P(B) = \frac{1}{2}$$

$$P(B|A) \neq P(B)$$

$\therefore A$ & B is not independent.

7. Three cards are drawn in succession (without replacement) from a 52 card deck. Find the probability that:

a) there are three aces

b) If the first is an ace, then the other two are aces

c) If the first two are aces, then the third is an ace.

let E_1 = First card is an ace

E_2 = Second card is an ace

E_3 = Third card is an ace

$$\begin{aligned} \text{a) } \therefore P(\text{all three aces}) &= P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3) \\ &= \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} \\ &= \frac{1}{13} \cdot \frac{1}{17} \cdot \frac{1}{25} \\ &= \frac{1}{5525}. \end{aligned}$$

(r)

$$P = \frac{{}^4C_3}{{}^{52}C_3} = \frac{4}{22100} = \frac{1}{5525}.$$

b). By conditional probability

$$P(E_2 \cap E_3 | E_1) = \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_1)}.$$

First is already an ace, find probability the next two are also aces.

$$\begin{aligned} &= \frac{\frac{1}{5525}}{\frac{4}{52}} \\ &= \frac{1}{5525} \times \frac{13}{1} = \frac{1}{425}. \end{aligned}$$

c) By conditional probability

$$P(E_3 | E_1, E_2) = \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_1 \cap E_2)}$$

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2) \\ = \frac{1}{13} \cdot \frac{1}{17} = \frac{1}{221}$$

$$\therefore P(E_3 | E_1, E_2) = \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_1 \cap E_2)} = \frac{1}{5525} \times \frac{221}{1} = \underline{\underline{\frac{1}{25}}}$$

First two are already
aces; probability
the third is an
ace.

8. Two marbles are selected one after the other with replacement from a box containing 3 white marbles & 2 red marbles.

Find the probability p that

a) The two marbles are white

b) The two marbles are red

c) The second is white if the first is white

d) The second is red if the first is red.

soln. Total marbles in the box = 3 white + 2 red = 5.
Selection with replacement $\Rightarrow$ Independent events.

$$P(W) = \frac{3}{5}, \quad P(R) = \frac{2}{5}$$

a) W_1 = event choosing white in first time
 W_2 = event choosing white in second time.

probability both are white

$$P(W_1 \cap W_2) = P(W_1) \cdot P(W_2)$$

$$= \frac{3}{5} \cdot \frac{3}{5} = \underline{\underline{\frac{9}{25}}}$$

$$P(W_1) = \frac{3}{5}, \quad P(W_2) = \frac{3}{5}$$

- b). R_1 = event choosing red in first time
 R_2 = event choosing red in second time.

$$P(R_1) = \frac{2}{5} \quad P(R_2) = \frac{2}{5}$$

probability both are red.

$$P(R_1 \cap R_2) = \frac{2}{5} \times \frac{2}{5} = \frac{4}{25} //$$

- c) probability second is white given first is white.
since replacement makes draws independent

$$P(W_2|W_1) = P(W_2)$$
$$= \frac{3}{5} // \quad (or)$$

$$P(W_2|W_1) = \frac{P(W_1 \cap W_2)}{P(W_1)} = \frac{\frac{9}{25}}{\frac{3}{5}} = \frac{9}{25} \times \frac{5}{3} = \frac{3}{5}.$$

- d). probability second is red given first is red
since replacement makes draws independent

$$P(R_2|R_1) = P(R_2)$$
$$= \frac{2}{5} // \quad (or)$$

$$P(R_2|R_1) = \frac{P(R_1 \cap R_2)}{P(R_1)} = \frac{\frac{4}{25}}{\frac{2}{5}} = \frac{4}{25} \times \frac{5}{2} = \frac{2}{5} //$$

A card is drawn at random from a well shuffled deck of 52 cards. Let A be the event that the card is a king & B be the event that the card is a Queen. Are A & B mutually exclusive? find $P(A)$, $P(B)$, $P(A \cap B)$.

Soln

Event A = "card is king"

Event B = "card is Queen"

One single card cannot be both king & Queen.
So events A & B cannot happen together.

i.e. $A \cap B = \phi$

$$P(A \cap B) = 0.$$

A & B are mutually exclusive.

• probability of king

Total no. of cards = 52

A deck has 4 kings.

$$P(A) = \frac{4}{52} = \frac{1}{13}.$$

• probability of Queen

Total no. of cards = 52

A deck has 4 Queens

$$P(B) = \frac{4}{52} = \underline{\underline{\frac{1}{13}}}.$$

Baye's theorem: An event A corresponds to a number of exhaustive events $B_1, B_2, B_3, \dots, B_n$.
 If $P(B_i)$ & $P(A/B_i)$ are given then

$$P(B_i/A) = \frac{P(B_i) \cdot P(A/B_i)}{\sum P(B_i) \cdot P(A/B_i)}$$

Theorem of total probability

Let $\{E_1, E_2, \dots, E_n\}$ be a partition of the sample space S & suppose that each of the events $E_1, E_2, \dots, E_n$ has non zero probability of occurrence. Let A be any event associated with S , then

$$\begin{aligned} P(A) &= P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + \dots + P(E_n) \cdot P(A/E_n) \\ &= \sum_{j=1}^n P(E_j) \cdot P(A/E_j). \end{aligned}$$

Problems.

1. A university bought 45%, 25%, & 30% of computers from HCL, WIPRO, IBM respectively. Supposed that 2% of computers from HCL, 3% of computers from Wipro & 1% of computers from IBM are found to be defective. Given a randomly chosen computer is defective, what is the probability that it is made by i) HCL ii) WIPRO?

$A \Rightarrow$ computer is Defective

$B_1 \Rightarrow$ computer made by HCL

$B_2 \Rightarrow$ computer made by WIPRO

$B_3 \Rightarrow$ computer made by IBM

$$P(B_1) = 0.45$$

$$P(B_2) = 0.25$$

$$P(B_3) = 0.30$$

$$P(A|B_1) = 0.02$$

$$P(A|B_2) = 0.03$$

$$P(A|B_3) = 0.01$$

Probability of Defective

$$\begin{aligned} P(A) &= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) \\ &= 0.45(0.02) + 0.25(0.03) + 0.30(0.01) \\ &= 0.0195 \end{aligned}$$

Applying the Baye's theorem

$$\begin{aligned} \text{i) } P(B_1|A) &= \frac{P(B_1) \cdot P(A|B_1)}{P(A)} \\ &= \frac{(0.45)(0.02)}{0.0195} \\ &= 0.4615 \end{aligned}$$

46.15% chance defective computer comes from HCL.

$$\begin{aligned} \text{ii) } P(B_2|A) &= \frac{P(B_2) \cdot P(A|B_2)}{P(A)} \\ &= \frac{(0.25)(0.03)}{0.0195} \\ &= 0.3846 \end{aligned}$$

38.46% chance defective computer comes from WIPRO.

2. A Consulting firm rents cars from three agencies such that 20% from Agency 'x', 30% from Agency 'y' & 50% from agency 'z'. 90% of the cars from 'x', 80% of cars from 'y' & 95% of the cars from 'z' are in good condition. If a car is in good condition, what is the probability that it has come from
 i) agency 'y' ii) agency 'z'.

Soln

$A \rightarrow$ car is in good condition
 $B_1 \rightarrow$ Car is from agency 'x'
 $B_2 \rightarrow$ car is from agency 'y'
 $B_3 \rightarrow$ car is from agency 'z'.

$$P(B_1) = 0.2$$

$$P(B_2) = 0.3$$

$$P(B_3) = 0.5$$

$$P(A|B_1) = 0.9$$

$$P(A|B_2) = 0.8$$

$$P(A|B_3) = 0.95$$

Total probability

$$\begin{aligned} P(A) &= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) \\ &= (0.2)(0.9) + (0.3)(0.8) + (0.5)(0.95) \\ &= 0.18 + 0.24 + 0.475 \\ &= 0.895 \end{aligned}$$

Apply Bay's rule

$$i) P(B_2|A) = \frac{P(B_2) \times P(A|B_2)}{P(A)} = \frac{(0.3)(0.8)}{0.895} = 0.2682$$

26.82% chance good condition car comes from agency 'y'.

$$\begin{aligned}
 \text{i) } P(B_3|A) &= \frac{P(B_3) \cdot P(A|B_3)}{P(A)} \\
 &= \frac{(0.5)(0.95)}{0.895} \\
 &= 0.5307\%
 \end{aligned}$$

53.07%. chance good condition car comes from agency Z.

3. police plan to enforce speed limits by using RADAR traps at 3 different locations within the city limits. The RADAR traps at each of the location 'P', 'Q', & 'R' are operated 40%, 30% & 20% of the time. A person who is speeding on her way to work has probabilities of 0.2, 0.1, & 0.5 respectively, of passing through these locations. If the person received a speeding ticket on her way to work, what is the probability that she passed through the RADAR trap located at i) location 'P' ii) location 'R'.

$A \rightarrow$ a person receives a speeding ticket
 $B_1 \rightarrow$ Speeding through location P
 $B_2 \rightarrow$ Speeding through location Q
 $B_3 \rightarrow$ Speeding through location R

$$\begin{aligned}
 P(B_1) &= 0.4 & P(A|B_1) &= 0.2 \\
 P(B_2) &= 0.3 & P(A|B_2) &= 0.1 \\
 P(B_3) &= 0.2 & P(A|B_3) &= 0.5
 \end{aligned}$$

Total probability

$$\begin{aligned}
 P(A) &= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) \\
 &= 0.211
 \end{aligned}$$

Apply Baye's theorem

$$i) P(B_1/A) = \frac{P(B_1) \cdot P(A/B_1)}{P(A)} = \frac{0.08}{0.2} = 0.3809 //$$

38.09% chance person received a speeding ticket from location P.

$$ii) P(B_2/A) = \frac{P(B_2) \cdot P(A/B_2)}{P(A)} = \frac{0.1}{0.21} = 0.4762 //$$

47.62% chance person received a speeding ticket from location R.

4. In a certain college, 4% of the boys & 1% of the girls are taller than 1.8m. Further more 60% of the students are girls. If a student is selected at random & is found to be taller than 1.8m, what is the probability that the student is a girl?

Soln:

A $\Rightarrow$ student found to be taller than 1.8m
 $B_1 \Rightarrow$ Boys taller than 1.8m Student is a boy
 $B_2 \Rightarrow$ Girls taller than 1.8m. Student is a girl

$$P(B_1) = \frac{40}{100} = 0.4$$

$$P(A/B_1) = 0.04$$

$$P(B_2) = \frac{60}{100} = 0.6$$

$$P(A/B_2) = 0.01$$

Total probability

$$\begin{aligned} P(A) &= P(B_1) \cdot P(A/B_1) + P(B_2) \cdot P(A/B_2) \\ &= 0.4(0.04) + (0.6)(0.01) \\ &= 0.022 // \end{aligned}$$

Apply Baye's theorem

$$P(B_2/A) = \frac{P(B_2) \cdot P(A/B_2)}{P(A)} = \frac{(0.6)(0.01)}{0.022} = 0.273$$

27.3% chance the student is a girl is taller than 1.8m. is a girl.

5. Three companies A, B, C supply 20%, 35% & 40% of the notebooks to a School respectively. Past experience shows that 5%, 4% & 2% of the notebooks produced by these companies are defective respectively. If a notebook was found to be defective, what is the probability that the notebook was supplied by company 'A'?

Soln :-

$A \rightarrow$ Notebook is defective
 $B_1 \rightarrow$ Notebook supplied by A
 $B_2 \rightarrow$ Notebook supplied by B
 $B_3 \rightarrow$ Notebook supplied by C.

$$P(B_1) = \frac{20}{100} = 0.2$$

$$P(B_2) = \frac{35}{100} = 0.35$$

$$P(B_3) = \frac{40}{100} = 0.4$$

$$P(A/B_1) = 0.05$$

$$P(A/B_2) = 0.04$$

$$P(A/B_3) = 0.02$$

Total probability

$$\begin{aligned}
 P(A) &= P(B_1) \cdot P(A/B_1) + P(B_2) \cdot P(A/B_2) + P(B_3) \cdot P(A/B_3) \\
 &= 0.2(0.05) + 0.35(0.04) + (0.4)(0.02) \\
 &= 0.032
 \end{aligned}$$

By Applying Bay's Theorem

$$\begin{aligned}
 P(B_1/A) &= \frac{P(B_1) \cdot P(A/B_1)}{P(A)} \\
 &= \frac{(0.2)0.05}{0.032} \\
 &= 0.3125
 \end{aligned}$$

31.25% chance that the notebook supplied by company 'A'

6. There are three bags: first containing 1 white, 2 red, 3 green balls; second 2 white, 3 red, 1 green ball & third 3 white, 1 red, 2 green balls. Two balls are drawn from a bag chosen at random. These are found to be one white & one red. Find the probability that the balls so drawn came from the second bag.

Soln

$A \Rightarrow$ The two balls are white & red

$B_1 \Rightarrow$ choosing Bag 1

$B_2 \Rightarrow$ choosing Bag 2

$B_3 \Rightarrow$ choosing Bag 3.

Now $P(B_1) = \frac{1}{3}$, $P(B_2) = \frac{1}{3}$, $P(B_3) = \frac{1}{3}$

From Bag 1

probability of choosing 1 white, 1 red.

$$P(A|B_1) = \frac{{}^1C_1 \times {}^2C_1}{{}^6C_2} = \frac{2}{15}$$

$${}^6C_2 = \frac{6 \times 5}{2}$$

From Bag 2

$$P(A|B_2) = \frac{{}^2C_1 \times {}^3C_1}{{}^6C_2} = \frac{2 \times 3}{15} = \frac{2}{5}$$

From Bag 3

$$P(A|B_3) = \frac{{}^3C_1 \times {}^1C_1}{{}^6C_2} = \frac{3 \times 1}{15} = \frac{1}{5}$$

Total probability

$$\begin{aligned} P(A) &= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) \\ &= \frac{1}{3} \times \frac{2}{15} + \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{1}{5} = \frac{11}{45} \end{aligned}$$

Apply Bayi's theorem

$$P(B_2|A) = \frac{P(B_2) \cdot P(A|B_2)}{P(A)}$$

$$= \frac{1}{3} \times \frac{2}{5} \times \frac{45}{11}$$

$$= \frac{6}{11} = 54.54\% \text{ } 0.5454$$

54.54% chance that the Balls were from Bag 2.

6. The Contents of three Urns are;
 1 white, 2 red, 3 green balls;
 2 white, 1 red, 1 green balls;
 4 white, 5 red, 3 green balls.

Two balls are drawn from an urn choosing at random. These are found to be one white & one green. Find the probability that the balls drawn came from this urn.

A $\rightarrow$ Choosing 1 white & 1 green.

B₁ $\rightarrow$ choosing urn 1 (1W, 2R, 3G)

B₂ $\rightarrow$ choosing urn 2 (2W, 1R, 1Gr)

B₃ $\rightarrow$ choosing urn 3 (4W, 5R, 3Gr).

$$P(B_1) = 1/3 \quad P(B_2) = 1/3 \quad P(B_3) = 1/3$$

i from urn 1

$$P(A|B_1) = \frac{{}^1C_1 \times {}^3C_1}{{}^6C_2} = \frac{3}{15} = 1/5$$

ii. from urn 2

$$P(A|B_2) = \frac{{}^2C_1 \times {}^1C_1}{{}^4C_2} = \frac{2/1}{6/1} = 1/3$$

from urn 3

$$P(A|B_3) = \frac{{}^4C_1 \times {}^3C_1}{{}^{12}C_2} = 2/11.$$

Total probability

$$\begin{aligned} P(A) &= P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3) \\ &= \frac{1}{3} \cdot \frac{1}{5} + \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{11} \\ &= \frac{1}{3} \left(\frac{1}{5} + \frac{1}{3} + \frac{2}{11} \right) \\ &= 0.237. \end{aligned}$$

By Applying Baye's theorem

$$\begin{aligned} P(B_3|A) &= \frac{P(B_3) \cdot P(A|B_3)}{P(A)} \\ &= \frac{\frac{1}{3} \cdot \frac{2}{11}}{0.237} = \frac{0.060}{0.237} \\ &= 0.253 // \end{aligned}$$